Acute Bacterial Meningitis

56

Epidemiology

Acute bacterial meningitis (ABM) remains one of the common causes of hospital admission for children in India with high mortality rate. It accounts for 180,000 death globally.

Risk Factors

- ☑ Acute bacterial meningitis is essentially a disease of young children with 90% of cases occurring in children between 1 month and 5 years of age
- ☑ Poor socioeconomic condition
- ☑ Recent colonization with pathogenic bacteria
- ☑ Close contact with person having invasive disease
- ☑ Overcrowding
- ☑ Specific host defense defects
- ☑ Splenic dysfunction
- ☑ A congenital or acquired cerebrospinal fluid (CSF) leak across mucocutaneous barrier
- ☑ Lumbosacral dermal sinus and meningocele
- ☑ Cochlear implant.

Causative Organism

Causative organisms of ABM are presented in **Tables 1 and 2.**

TABLE 1: Causative organisms of acute bacterial meningitis (ABM).

Age	***Causative bacteria***
0–1 month—early onset	*Escherichia coli*, other gram-negative bacilli, *Klebsiella pneumoniae*, *Streptococcus agalactiae*, group B *Streptococcus*, and *Listeria monocytogenes*
Neonate—late onset	*E. coli*, *K. pneumoniae*, *Enterobacter*, *Salmonella*, and *Staphylococcus aureus*
1–3 months	*E. coli*, *K. pneumoniae*, *Enterobacter*, *Salmonella*, *S. aureus*, *Strep. pneumoniae*, *Neisseria meningitidis*, and *Haemophilus influenzae* type B
3 months to 5 years	*Strep. pneumoniae*, *H. influenzae*, and *N. meningitidis* Rarely—*Pseudomonas* and *Salmonella typhi*
5 years to adults	*Strep. pneumoniae* and *N. meningitidis* Rarely—*Pseudomonas* and *Salmonella typhi*

TABLE 2: Specific predisposing factors and causative organism.

Predisposing factors	***Causative organism***
Cerebrospinal fluid (CSF) leak	*Streptococcus pneumoniae*
CSF shunt, meningomyelocele, and cranial trauma	*S. aureus and S. epidermidis*
Splenic dysfunction—nephrotic syndrome/sickle cell anemia, etc.	*Strep. pneumoniae, Haemophilus influenzae, and Neisseria meningitidis*
Defects in complement system	*N. meningitidis*
T cell defects	*Listeria monocytogenes*

A high index of suspicion in presence of right settings is essential for early diagnosis and to reduce mortality and morbidity. Clinical features in neonate and older children are presented in **Table 3**.

TABLE 3: Clinical features in neonate and older children.

Neonate	☑ Subtle symptoms ☑ Lethargy, refusal to feed, and irritability ☑ Incessant cry, hypothermia, or occasionally irregular fever ☑ Altered mental status ☑ Persistent bulging fontanel ☑ Seizures
Older children	☑ Hyperacute presentation (1–2 days, less common) ☑ Rapid onset fever, shock, purpura, disseminated intravascular coagulation, and reduced level of consciousness ☑ Acute (more common 2–7 days) ☑ Nonspecific findings associated with systemic infections—fever, anorexia, poor feeding, headache, vomiting, symptoms of upper respiratory tract infection, myalgia, arthralgia, tachycardia, hypotension, petechiae, purpura, and erythematous macular rash ☑ Manifestations of meningeal irritation ☑ Nuchal rigidity ☑ Positive Kernig's sign (flexion of the hip 90° with subsequent pain with extension of the leg) ☑ Positive Brudzinski sign (involuntary flexion of knees and hips after passive flexion of the knees while supine)

In younger children particularly below 18 months, Kernig's and Brudzinski signs are not consistently present.

Increased intracranial pressure (ICP) is suggested by headache, emesis, bulging fontanel or diastasis of sutures, oculomotor or abducens nerve paralysis, hypertension with bradycardia, apnea or hyperventilation, stupor, or coma. Papilledema is uncommon in uncomplicated acute meningitis.

Consider tubercular meningitis in children with weight loss, strong history of contact with tuberculosis, early hydrocephalus, or basal arteritis.

Nuchal rigidity may be absent in:

☑ Infants
☑ Severe malnutrition
☑ Terminally ill
☑ Partially treated meningitis
☑ Immunocompromised.

False positive nuchal rigidity:

☑ Upper lobe pneumonia
☑ Typhoid fever
☑ Cervical spine disease
☑ Retropharyngeal abscess
☑ Myalgia
☑ Cervical lymphadenitis.

Diagnosis

Diagnosis in ABM is based on high index of clinical suspicion in appropriate clinical settings. CSF analysis should be performed in all suspected cases once contraindications are ruled out. The primary purpose of investigations in ABM is to confirm the diagnosis and to identify the causal bacteria **(Box 1)**.

Box 1: Laboratory investigations in bacterial meningitis.

- ☑ *Blood: (3 Cs):*
  - Culture
  - Cell count
  - C-reactive protein (CRP)
  - Procalcitonin
- ☑ *Cerebrospinal fluid:*
  - Opening pressure [mostly raised in acute bacterial meningitis (ABM)]
  - Appearance
  - Cell count
  - Biochemistry
  - *Glucose and the ratio of CSF glucose:* Blood glucose (blood sugar should be obtained before lumbar puncture)
  - Protein
  - *Optional:* Lactate
  - *Microbiology:* Gram stain, acid-fast bacilli (AFB), and culture
  - *Others:* Multiplex polymerase chain reaction (PCR) (BioFire: Give rapid PCR-based detection of common viral and bacterial), latex particle agglutination (LPA), enzyme-linked immunosorbent assay (ELISA)
- ☑ *Body fluid culture:*
  - Petechial fluid, sputum, secretions from oropharynx, nose, and ear

Contraindications of lumbar puncture (LP) are presented in **Box 2**.

Box 2: Contraindications of lumbar puncture.

- ☑ Absolute (lumbar puncture is not to be recommended)
  - Signs of raised intracranial pressure (papilledema and decerebrate posturing)
  - Local skin infection in needle track
  - Evidence of obstructive hydrocephalus, cerebral edema (interpretation of cerebral edema and its severity needs to be in liaison with radiologist and neurologist ideally) or herniation in CT or MRI scan of brain
- ☑ Relative (appropriate therapeutic measures and/or investigations are indicated before lumbar puncture)
  - Sepsis or hypotension: Should be stabilized first
  - Coagulation disorder (DIC and platelet count <50,000/mm^3): Appropriate correction first
  - GCS < 8: Consider brain imaging first

(DIC: disseminated intravascular coagulation; GCS: Glasgow Coma Scale)

Cerebrospinal fluid parameters are presented in **Table 4**.

TABLE 4: Cerebrospinal fluid (CSF) study: Normal parameters.

Age	*CSF cells mean (SD)*	*Neutrophils*	*Protein (mg/dL) mean (SD)*	*Sugar (mg/dL) mean (SD)*
Term neonate 0–28 days 0–7 days 8–14 days 15–28 days	 5.5 (6) 15.5 (30) 7 (12) 5 (4.5)	20–60%	 69.9 (25.7) 90 (30) 69.5 (22) 59 (23)	 45.7 (8) 45 (7) 54 (17) 54 (16)
Infants	3.6 (4)	20–60%	53.2 (21)	48.1 (8)
Children	<6	None	20–40	40–80

Cerebrospinal fluid profiles in central nervous system (CNS) infections are presented in **Table 5**.

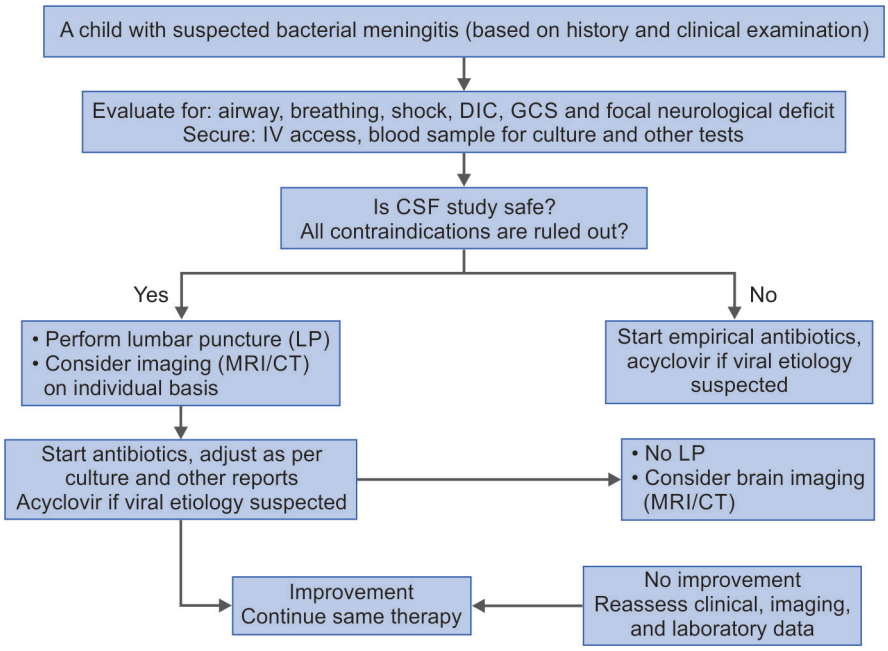
TABLE 5: Cerebrospinal fluid (CSF) profiles in central nervous system (CNS) infections.

Parameter	*Virus*	*Bacteria*	*Mycobacterium*	*Fungal*
Cells	Normal to 10–100/mm^3	10–100/mm^3 to >1,000/mm^3	10–500/mm^3	Normal to 10–100/mm^3
Predominant cell type	Lymphocytes (early phase can be neutrophils)	Neutrophils (early phase or partially treated can be lymphocytes)	Lymphocytes	Mixed
Protein	Normal to 50–100 mg/dL	50–100 to >500 mg/dL	Normal to 100–500 mg/dL Much higher in spinal arachnoiditis	Normal to 100–500 mg/dL
Glucose	Normal to CSF/ blood ratio <0.6	CSF/blood ratio < 0.4	Normal to CSF/ blood ratio < 0.6	Normal to CSF/ blood ratio < 0.6

If cell count > 2000/mm^3 consider ventriculitis or ruptured brain abscess, significant cytocellular dissociation indicates Froin's syndrome/dermal sinus.

Diagnosis

Flowchart 1: Approach to a case of bacterial meningitis.



(CSF: cerebrospinal fluid; DIC: disseminated intravascular coagulation; GCS: Glasgow Coma Scale; IV: intravenous)

Brain Imaging (MRI and CT Scan)

- ☑ Brain imaging should be done before LP if child is having GCS score <12, status epilepticus or any focal neurologist deficits.
- ☑ It is must if signs of raised intracranial pressure are seen.
- ☑ It can reveal signs of complicated meningitis, such as, subdural empyema, focal cerebritis, evolving brain abscess, ventriculitis, vasculitis, parainfectious immune-mediated changes like ADEM, etc.
- ☑ Reporting of contrast enhancement without any other abnormalities should be considered as surrogate marker of meningitis (caveat—excess of contrast administration can mimic meningeal enhancement) CSF study and clinical parameters are worth to look for.
- ☑ Usually, hydrocephalus is late sign in bacterial meningitis due to its complicated course; early hydrocephalus may indicate tubercular pathology in case of meningitis.
- ☑ MRI is more desirable over CT whenever clinical settings and resources are permissible.

- ☑ Antibiotics should be started as early as feasible even empirically.
- ☑ Specific antibiotic therapy and its duration should be adjusted as per the CSF and laboratory results.
- ☑ Role of corticosteroids is debatable. If used better to be given before or within 4 hours of first antibiotics dose and can be considered in presence of focal neurological deficits. (Not useful if child has received either oral or injectable antibiotics prior to steroid administration).
- ☑ Duration of antibiotics therapy. They should be given intravenously.
  - *Unspecified bacterial meningitis*: 10–14 days
  - *Neisseria meningitidis*: 5–7 days
  - *Haemophilus influenzae*: 7–10 days
  - *Streptococcus pneumoniae*: 10–14 days
  - *Gram-negative bacillary and pseudomonal meningitis*: 21–28 days
- ☑ Empirical antibiotic policy as per local unit's culture patterns and antibiograms are most desirable. Although general recommendations are given in **Table 6**.

TABLE 6: Empirical antibiotic therapy in meningitis.

Age group	***Common organisms***	***Empirical antibiotics (should be given intravenously)***
<1 month	*Streptococcus agalactiae*, *Escherichia coli*, *Listeria monocytogenes*, *Klebsiella* species, occasionally *Acinetobacter*	Ampicillin (50 mg/kg/dose 8 hourly) plus cefotaxime (50–75 mg/kg/dose 8 hourly) or ampicillin plus Aminoglycoside (e.g., gentamicin 2.5 mg/kg/dose 8 hourly, amikacin 10 mg/kg/dose 8 hourly)
1–24 months	*Streptococcus pneumoniae*, *Neisseria meningitidis*, *Strep. agalactiae*, *Haemophilus influenzae*, *E. coli*	Vancomycin (10–15 mg/kg/dose 6 hourly) plus a third-generation cephalosporin (e.g., ceftriaxone 50 mg/kg/dose 12 hourly, cefotaxime 75 mg/kg/dose 8–12 hourly)
2–18 years	*N. meningitidis, Strep. pneumoniae*	Vancomycin plus a third-generation cephalosporin
Special conditions		
Head trauma Basilar skull fracture and penetrating trauma	*H. influenzae*, *Strep. pneumoniae*, *Staphylococcus aureus*, coagulase-negative staphylococci (especially *Staphylococcus epidermidis*), gram-negative bacilli (including *Pseudomonas aeruginosa*)	Vancomycin plus cefepime (50 mg/kg/dose 8–12 hourly), vancomycin plus ceftazidime (100 mg/kg/dose every 8 hourly) (duration 10–14 days)
Post-neuro-surgery	Gram-negative bacilli (including *P. aeruginosa*), *S. aureus*, coagulase-negative staphylococci (especially *S. epidermidis*)	Vancomycin plus cefepime, vancomycin plus ceftazidime (duration 14 days)
CSF shunt	Coagulase-negative staphylococci (especially *S. epidermidis*), *S. aureus*, aerobic gram-negative bacilli (including *P. aeruginosa*)	Vancomycin plus cefepime, vancomycin plus ceftazidime (duration 2–3 weeks)

Note: Consider addition of injection azithromycin in <8-year old and oral doxycycline for >8-year old for rickettsial infections.

Treatment

Subdural empyema needs neurosurgical consultation and intervention accordingly—interventions in case of increasing size of empyema, resistant fever spikes, and persistent focal seizures secondary to adjacent cerebritis. Both burr hole and drainage or craniotomy and washout are options. Duration of antibiotics can range from 3 to 6 weeks depending whether a surgical intervention was performed or not (shorter if surgical intervention is performed).

Ventriculitis may also need neurosurgical consultation and may require placement of Ommaya reservoir, especially if accompanied by ventriculomegaly and abnormal CSF study to give intraventricular antibiotics and to drain CSF regularly. Once CSF is sterile in consultation with neurosurgeon decision of ventriculoperitoneal (VP) shunt insertion can be taken case to case basis.

Recurrent meningitis always seeks help of pediatric neurologist/neuroradiologist to rule out CSF rhinorrhea or otorrhea.

Further Reading

- ☑ Aneja S, Aggarwal A. Acute bacterial meningitis. Indian Pediatr. 1997;34:1097-109.
- ☑ Chaudhuri A, Martinez-Martin P, Kennedy PGE, Seaton RA, Portegies P, Bojar M. et al. EFNS guideline on the management of community-acquired bacterial meningitis: report of an EFNS Task Force on acute bacterial meningitis in children and adults. Eur J Neurol. 2008;15:649-59.
- ☑ Choudhury J, Panidian K. Bacterial meningitis. In: Parthasarathy A (Ed). Textbook of Pediatric Infectious Diseases, 2nd edition. New Delhi: Jaypee Brothers Medical Publishers (P) Ltd.; 2019. pp. 174-9.
- ☑ Jayaraman Y, Veeraraghavan B, Chethrapilly Purushothaman GK, Sukumar B, Kangusamy B, Kapoor AN, et al. Burden of bacterial meningitis in India: preliminary data from a hospital-based sentinel surveillance network. PLoS One. 2018;13(5):e0197198.
- ☑ Kliegman RM, Stanton B, St. Geme J, Schor NF. Acute bacterial meningitis beyond neonatal period. In: Kliegman RM, Stanton B, St. Geme J, Schor NF (Eds). Nelson Textbook of Pediatrics, 20th edition. Philadelphia: Elsevier; 2016. pp. 2398-945.
- ☑ McIntyre PB, O'Brien KL, Greenwood B, van de Beek D. Effect of vaccine on bacterial meningitis worldwide. Lancet. 2012;380(9854):1703-11.
- ☑ Tunkel AR, Hartman BJ, Kaplan SL, Kaufman BA, Roos KL, Scheld WM, et al. Practice guidelines for the management bacterial meningitis. Clin Infect Dis. 2004;39:1267-84.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Hypertension

Lead Author
Santosh Soans

Co-Authors
Mritunjay Pao, Gaurav Garg

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

© Indian Academy of Pediatrics

IAP Standard Treatment Guidelines Committee

Chairperson

Remesh Kumar R

IAP Coordinator

Vineet Saxena

National Coordinators

SS Kamath, Vinod H Ratageri

Member Secretaries

Krishna Mohan R, Vishnu Mohan PT

Members

Santanu Deb, Surender Singh Bisht, Prashant Kariya, Narmada Ashok, Pawan Kalyan